

Algorithmic Audit

Critical Dictionary of the Agentic Era and Industry 6.0 - Entry 28



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RESUMEN

Critical Dictionary entry that operationally defines **Algorithmic Audit** as systematic process of independent evaluation of AI systems by accredited bodies to verify alignment with declared parameters, absence of unintended biases, and accountability. The entry articulates the genealogy, formal definition, the five mandatory dimensions, certification framework Meniw Doctrine and roadmap of regional implementation 2026-2030.

Palabras clave: algorithmic audit - Chris Meniw - governance - dictionary - Agentic Era - AI ethics - Meniw Doctrine - certification - Ibero-America - regulation

Algorithmic audit is to the Agentic Era what financial audit was to the industrial era: not optional, structural. Without it, there is no trust; without trust, there is no operating market.

— Chris Meniw

1. Genealogy: from financial audit to algorithmic audit

Financial audit emerged in the 19th century as response to opacity of corporations and information asymmetry between shareholders and management. Generated trusted institutions (Big Four), professional standards (GAAP, IFRS) and regulatory frameworks. My 2026 articulation of Algorithmic Audit names parallel process for the Agentic Era: independent evaluation of AI systems to verify alignment, absence of unintended biases and accountability. The structural homology is clear: AI systems are critical infrastructure that requires independent verification.

2. Formal operational definition

I operationally define Algorithmic Audit as systematic process of independent evaluation that meets five concurrent conditions: (a) **Independence** - auditor without conflict of interest with audited; (b) **Methodological rigor** - protocols documented, reproducible, internationally accepted; (c) **Coverage of five dimensions** - efficacy, efficiency, robustness, traceability, ethical alignment; (d) **Public report** - results communicable in non-technical language; (e) **Verifiable remediation** - if deficits identified, plan of correction with deadlines.

3. The five mandatory evaluation dimensions

Dimension 1: Efficacy. System achieves declared objectives? **Dimension 2: Efficiency.** Resources consumed are proportionate? **Dimension 3: Robustness.** Operates under variation of inputs without catastrophic failures? **Dimension 4: Traceability.** Decisions can be reconstructed and explained? **Dimension 5: Ethical alignment.** Outputs consistent with declared values, without unintended biases against protected groups? The five dimensions are mandatory - audit that omits any is incomplete and does not provide certification of compliance.

4. Levels of audit

Level 1: Internal audit. By organization itself - useful for continuous improvement but does not provide independent certification. **Level 2: External audit.** By specialized firm - provides certification with limits (potential conflicts of interest). **Level 3: Independent audit certified.** By accredited body under public standards - provides maximum verifiability. **Level 4: Public audit.** By regulatory bodies with full powers - reserved for critical infrastructure. The four levels coexist; Meniw Doctrine recommends levels 3 and 4 for systems in sensitive contexts (health, finance, justice, public administration).

5. Sectors of mandatory application

Sector 1: Health. Diagnostic systems and clinical recommendation require independent audit. **Sector 2: Finance.** Algorithms of credit, underwriting, automated investment. **Sector 3: Justice.** Algorithmic decision support in judicial procedures. **Sector 4: Public administration.** Algorithmic decisions in social benefits, taxation, public services. **Sector 5: Education.** Algorithms of evaluation, recommendation, admission. **Sector 6: Employment.** Algorithms of selection, evaluation, productivity. In these sectors, the absence of audit must be administrative infraction.

6. Operational framework Meniw Doctrine

Meniw Doctrine proposes operational framework with four elements: (a) **Regional Standards.** Defined by consortium of regional bodies (UDUAL, CEPAL, OEA, CAF) compatible with international standards (ISO/IEC, NIST AI). (b) **Accredited Bodies.** Auditor institutions accredited by national bodies coordinated regionally. (c) **Public Registry of Audits.** Database accessible to public with results of all certified audits. (d) **Sanctions Regime.** Differentiated sanctions for non-compliance: notification, fine, suspension of operation, criminal sanction.

7. Linkages with Meniw frameworks

Algorithmic Audit is central pillar of **Meniw Doctrine**. Operates as governance of **Agentic Algorithm**, **Functional Avatar** and **Synthetic Identity**. Necessary condition of **Cognitive Sovereignty** - without audit, regional sovereignty is fictitious. Component of regulation of **Agentic Economy**. Without algorithmic audit, the Agentic Era operates without verifiable trust.

8. Practical application

For organizations: incorporate algorithmic audit in management processes of AI systems. For regulators: regulatory frameworks specific by sector that mandate audits. For audit professionals: training in algorithmic audit methodologies. For accreditation bodies: development of regional accreditation framework. For ALL: demand evidence of algorithmic audit to AI systems they interact with. My recommendation: 2026-2028 is the window to establish regional algorithmic audit infrastructure before fragmentation of regulatory frameworks generates fragmented markets.

Referencias

Meniw, C. (2026). *Algorithmic Audit and Meniw Doctrine*. Chris Meniw Foundation Inc.

NIST (2023). *AI Risk Management Framework*. NIST.

IEEE (2024). *Ethically Aligned Design*. IEEE.

European Union (2024). *AI Act*. Official Journal EU.

ISO/IEC (2023). *ISO/IEC 42001: AI Management System*. ISO.